90. Consider the following

1. Photosynthesis
2. Respiration
3. Decay of organic matter
4. Volcanic action

Which of the above add carbon dioxide to the carbon cycle on Earth?

(a) 1and 4 only (b) 2 and 3 only
(c) 2, 3 and 4 (d) 1, 2, 3 and 4

Ans. (c) : Carbon dioxide is a colourless, odourless gas. Carbon dioxide in the atmosphere is found to be 0.03 percent as required. Its aqueous solution is acidic. At atmospheric pressure, it transforms into solid -state at 78°C temperature, which is called 'dry ice'. 'Dry ice' is used in refrigeration, respiration, weathering of biomaterials and volcanic activity adds carbon dioxide to the carbon cycle while photosynthesis, weathering of rocks etc., results in reduced carbon dioxide in the atmosphere.

91. Biodiversity forms the basis for human existence in the following ways :

1. Soil formation
2. Prevention of soil erosion
3. Recycling of waste
4. Pollination of crops

Select the correct answer using the code given below :

(a) 1, 2 and 3 only (b) 2, 3 and 4 only
(c) 1 and 4 only (d) 1, 2, 3 and 4

Ans. (d): Biodiversity remains the basis of human existence through soil formation, soil erosion prevention, crop pollination and recycling of waste.

92. Which one of the following is not a site for in-situ method of conservation of flora?

(a) Biosphere Reserve (b) Botanical Garden
(c) National Park (d) Wild life Sanctuary

Ans. (b): Botanical garden does not come under the in- situ method of conservation of flora, while the biosphere reserve, national park and wildlife sanctuary are included in it.

CURRENT AFFAIRS

93. Recently, the USA decided to support India's membership in multi-lateral export control regimes called the "Australia Group" and the "Wassenaar Arrangement". What is the difference between them ?

1. The Australia Group is an informal arrangement which aims to allow exporting countries to minimize the risk of assisting chemical and biological weapons proliferation, whereas the Wassenaar Arrangement is a formal group under the OECD holding identical objectives.
2. The Australia group comprises predominantly of Asian, African and North American countries, whereas the members countries of Wassenaar Arrangement are predominantly from the European Union and American Continents.

Which of the statements given above is/ are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d): The Australia Group (AG) is an informal association that works on the basis of consensus. It aims to allow exporters or transshipment countries to minimize the risk of further proliferation of Chemical and Biological Weapons (CBW). There are not many Asian/African 'countries in the Australian group. There are 43 members, including India, in this group.

94. In the context of global oil prices, ''Brent crude oil'' is frequently referred to in the news. What does this term imply?

1. It is a major classification of crude oil.
2. It is sourced from North Sea.
3. It does not contain sulphur.

Which of the statements given above is/are correct?

(a) 2 only (b) 1 and 2 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (b): In the context of global oil prices, 'Brent crude oil' is often mentioned is the news. Brent crude oil is the largest of the major classification of crude oil. It is obtained from the North Sea. This includes Brent crude, Brent sweet, Light crude, Ecofisk and Oseberg. This oil is used for pricing about two-thirds of the world's international trade of crude oil.

95. Recently,'' oilzapper'' was in the news. What is it?

(a) It is an eco-friendly technology for the remediation of oily sludge and oil spills
(b) It is the latest technology developed for under sea oil exploration
(c) It is a generically engineered high biofuel yielding maize variety
(d) It is the latest technology to control the accidentally caused flames oil wells.

Ans. (a) : Recently, 'oilzapper' was in the news. It is an ecologically developed technology for the treatment of oily punk and scattered oil. It has been developed by 'The Energy and Research Institute (TERI).

MISCELLANEOUS

96. The new START treaty was in the news. What is this treaty ?

(a) It is a bilateral strategic nuclear arms reduction treaty between the USA and the Russian Federation.
(b) It is a multilateral energy security cooperation treaty among the members of the East Asia Summit.
(c) It is a treaty between the Russian Federation and the European Union for the energy security cooperation.
(d) It is a multilateral cooperation treaty among the BRICS countries for the promotion of trade.

Ans. (a) : The New Strategic Arms Reduction Treaty (New START) was signed in 2010 in Prague by Russia and the United States. This treaty continues the bipartisan process of verifiably reducing US and Russian strategic nuclear arsenals begun by former Presidents Ronald Reagan and George H.W. Bush through START-I in 1994.

97. Karl Marx explained the process of class struggle with the help of which one of the following theories ?

(a) Empirical liberalism
(b) Existentialism
(c) Darwin's theory of evolution
(d) Dialectical meterisalism

Ans. (d) : Darwin's 'struggle for existence' in natural history is analogous to the class struggle in "human history". So option (c) is incorrect. The sketch of dialectical materialism given by Karl Marx explains that Dialectical principle is of universal application and all the development is "struggle of opposites". Thus option (d) is the correct answer.

98. With reference to Look East Policy of India, consider the following statements :

1. India wants to establish itself as an important regional player in the East Asian Affairs
2. India Wants to plug the vacuum created by the termination of cold War.
3. India wants to restore the historical and cultural ties with its neighbors in Southeast and East Asia.

Which of the statements given above is/are correct?

(a) 1 only (b) 1 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans. (b) : The main objective of the look east policy is economic integration with east and southeast Asia. Indian realized that it's East Asian neighbours achieved rapid economic growth and that it was lagging behind.

99. India maintained its early cultural contacts and trade links with Southeast Asia across the Bay of Bengal. For this pre-eminence of early maritime history of Bay of Bengal, which of the following could be the most convincing explanation/ explanations?

(a) As compared to other countries, India had a better ship-building technology in ancient and medieval times.
(b) The rulers of southern India always patronized traders,Brahmin priests and Buddhist monks in this context.
(c) Monsoon winds across the Bay of Bengals Facilitated sea voyages.
(d) Both (a) and (b) are convincing explanations in this context.

Ans. (c): India had maintained its initial cultural contacts and trade relations with South-East Asia even beyond the Bay of Bengal. The main reason for this was the ease of sea voyages through the monsoon winds in the Bay of Bengal.

100. With reference to the period of colonial rule in India,'' Home Charges' formed an important part of drain of wealth from India. Which of the following funds constituted Home Charges?

1. Funds used to support the India Office in London
2. Funds used to pay salaries and pensions of British personnel engaged in India
3. Funds used for waging wars outside India by the British

Select the correct answer using the codes given below :

(a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans. (b) : In India during the colonial rule, "Home Charges" was an important part of the exploitation of property in India. In Colonial India, Home Charges was the expenditure incurred by the secretary of state in England on behalf of India. This included the India office in London. Funds used for maintenance, salaries of English employees working in India and funds used for pension, etc., were included.